CARD 1 OF 8 DURATION **QUESTION**

Hydraulically calculated system: how long must the water supply last for **light, ordinary and extra hazard**?

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CARD 2 OF 8 LOW END OF THE RANGE **QUESTION**

Ordinary hazard says **60–90 minutes**. What earns you the **60**?

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CARD 3 OF 8 HOSE STREAM ALLOWANCE **QUESTION**

What is the **total hose stream allowance** (inside + outside) for light, ordinary and extra hazard, hydraulically calculated?

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CARD 4 OF 8 INSIDE HOSE **QUESTION**

The building has inside hose connections off the sprinkler system. How much flow do you add, and **where** in the calc?

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*FSM · Sprinkler Fact Cards — NFPA 13 (2019) Water Supply Duration & Hose Streams · sheet 1 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

CARD 2 OF 8 LOW END OF THE RANGE **ANSWER**

Monitored alarms. The waterflow alarm and supervisory devices must be **electrically supervised**, and that supervision must report to an approved, **constantly attended** location.

No monitoring → design to the **high** end (90).

NFPA 13 (2019) · Ch. 19

On the job: a local bell on the outside wall with no central station doesn't count. Ask how the alarm panel reports before picking the low number.

CARD 1 OF 8 DURATION **ANSWER**

Light hazard: 30 minutes.
Ordinary hazard: 60–90 minutes.
Extra hazard: 90–120 minutes.

The supply has to deliver the sprinkler demand **plus** the hose stream allowance for that whole time.

NFPA 13 (2019) · Ch. 19

Trap: storage. A rack warehouse doesn't use this table. Chapters 20–25 set their own hose and duration numbers.

CARD 4 OF 8 INSIDE HOSE **ANSWER**

One hose connection: 50 gpm.
More than one: 100 gpm total.

Where: in 50 gpm steps, starting at the **most remote** hose connection, each at the pressure the sprinkler calc needs at that point.

Combined sprinkler/standpipe (Class I or III) in a fully sprinklered building: no inside hose demand.

NFPA 13 (2019) · Ch. 19

Trap: dumping all 100 gpm at the base of the riser. It goes in at the hose connections.

CARD 3 OF 8 HOSE STREAM ALLOWANCE **ANSWER**

Light hazard: 100 gpm.
Ordinary hazard: 250 gpm.
Extra hazard: 500 gpm.

Inside hose portion for each: 0, 50 or 100 gpm. It comes out of that total; it isn't added on top of it.

NFPA 13 (2019) · Ch. 19

Trap: adding 250 gpm of hose and then another 100 gpm of inside hose on an OH job. The 250 is the combined total.

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 OUTSIDE HOSE **QUESTION**

Where in the hydraulic calc does the **outside hose** allowance get added?

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CARD 6 OF 8 PIPE SCHEDULE **QUESTION**

On a **pipe schedule** system, what pressure, flow and duration does NFPA 13 require for light and ordinary hazard?

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CARD 7 OF 8 PIPE SCHEDULE LOW NUMBERS **QUESTION**

Pipe schedule system: what lets you use the **lower flow**, and what lets you use the **lower duration**?

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CARD 8 OF 8 MIXED HAZARDS **QUESTION**

One system covers an **office (light)** and a **shop (ordinary)**. Whose hose allowance and duration do you use?

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CARD 6 OF 8 PIPE SCHEDULE **ANSWER**

Light hazard: 15 psi residual · 500–750 gpm · 30–60 min.
Ordinary hazard: 20 psi · 850–1500 gpm · 60–90 min.

Flow is measured at the **base of the riser** and includes hose. The residual pressure is needed at the **highest sprinkler**.

Extra hazard: no pipe schedule numbers. It gets hydraulic calcs.

NFPA 13 (2019) · Ch. 19

Trap: quoting 30 minutes for light hazard on an old pipe schedule system. The pipe schedule range is 30–60.

CARD 5 OF 8 OUTSIDE HOSE **ANSWER**

At the city main connection or a private hydrant, whichever is closer to the system riser.

Example: the private hydrant is 80 ft from the riser and the city tap is 200 ft away. The outside hose goes in at the hydrant.

NFPA 13 (2019) · Ch. 19

Trap: adding it at the base of the riser. It belongs at the supply end of the underground.

CARD 8 OF 8 MIXED HAZARDS **ANSWER**

Two options:
• Use the **highest** hazard's numbers for the whole system, or
• Use **each** hazard's numbers in the calc for that hazard's design area.

Small-room shortcut: if the higher hazard is only in single rooms of **400 sq ft or less**, none of them next to each other, the rest of the system can use the main occupancy's numbers.

NFPA 13 (2019) · Ch. 19

On the job: one 300 sq ft mechanical room in an office building doesn't make the whole system ordinary hazard.

CARD 7 OF 8 PIPE SCHEDULE LOW NUMBERS **ANSWER**

Lower duration: same as hydraulic. The waterflow and supervisory devices are electrically supervised and monitored at a constantly attended location.

Lower flow: the building is **noncombustible**, or no open area is bigger than **3000 sq ft** (light hazard) or **4000 sq ft** (ordinary hazard).

NFPA 13 (2019) · Ch. 19

Trap: assuming one condition unlocks both. Flow and duration have separate tests.

sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back